Exp - 6

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Error detection and correction using Hamming Code

Aim:- To implement error detection and correction using Hamming code

Description:-

* It adds redundant (parity) bits to the data so the receiver can find and correct single-bit error.

* parity bits positions → 1, 2, 4, 8, 16, ... (power of 2)
for 8 bit data, 8 + 4 = 12 bits.

* Parity bits work on even parity

$P_1 = 1, 3, 5, 9, 11$ $P_2 = 2, 3, 6, 7, 10, 11$

$P_4 = 4, 5, 6, 7, 12$ $P_5 = 8, 9, 10, 11, 12$

* At receiver, parity is recomputed.
If result = 0, no error.

Algorithm:-

- Sender: Reads 8 bit binary input. Data bits are placed where position ≠ 2^n. codeword is in "channel".
- Receiver: Reads "channel", recomputes parity. If non-zero, display error position and flipit and give original data.

Program Code:-

(i) Sender.py

```
val = input("Enter 8 bits: ").strip()
d = [int(x) for x in val]
p1 = d[0] ^ d[1] ^ d[3] ^ d[4] ^ d[6]
p2 = d[0] ^ d[2] ^ d[3] ^ d[5] ^ d[6]
p4 = d[1] ^ d[2] ^ d[3] ^ d[7]
p8 = d[4] ^ d[5] ^ d[6] ^ d[7]
code = [p1, p2, d[0], p4, d[1], d[2], d[3], p8,
        d[4], d[5], d[6], d[7]]
with open("channel", "w") as f:
    f.write("".join(map(str, code)))
print("Sent:", "".join(map(str, code)))
```

(ii) receiver.py

```
with open("channel", "r") as f:
    b = [int(x) for x in f.read().strip()]
s1 = b[0] ^ b[2] ^ b[4] ^ b[6] ^ b[8] ^ b[10]
s2 = b[1] ^ b[2] ^ b[5] ^ b[6] ^ b[9] ^ b[10]
s4 = b[3] ^ b[4] ^ b[5] ^ b[6] ^ b[11]
s8 = b[7] ^ b[8] ^ b[9] ^ b[10] ^ b[11]
pos = s1*1 + s2*2 + s4*4 + s8*8
print("Received: ", "".join(map(str, b)))
```

```
if (pos==0):
    print("No Flip")
else:
    print("Error Detected")
    print(f"Position: {pos}")
    b[pos-1]^=1
    print("Fixed:","".join(map(str,b)))

data=[b[2],b[4],b[5],b[6],b[8],b[9],b[10],b[11]]
print("Decoded Output:","".join(map(str,data)))
```

Input:-

10110010

Channel:-

10100111100 10

if bit flipped,

Channel:-

10101111 00 10

Output:-

Error Detected

Position: 5

fixed: 10100111 00 10

Decoded Output: 10110010

Result:-

The hamming code program detected the position of single-bit error, corrected it, & recovered the original message.

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